

Администрирование локальных сетей

Лабораторная работа №4

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Цели и задачи работы

Настроить взаимодействие через сеть провайдера посредством статической маршрутизации локальной сети организации с сетью основного здания, расположенного в 42-м квартале в Москве, и сетью филиала, расположенного в г. Сочи.

Ход выполнения работы

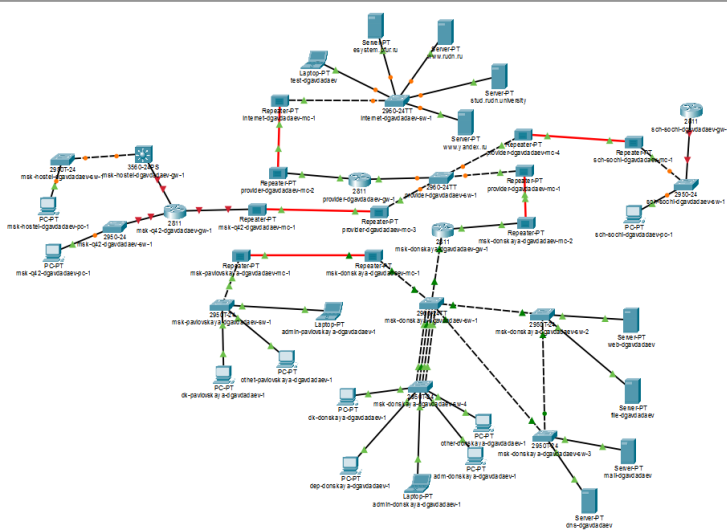


Рис. 1: Топология сети в Packet Tracer

Настройка сети провайдера

```
provider-dgavdadaev-sw-1#
provider-dgavdadaev-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
provider-dgavdadaev-sw-1(config)#int f0/3
provider-dgavdadaev-sw-1(config-if)#sw mode trunk

provider-dgavdadaev-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

provider-dgavdadaev-sw-1(config-if)#int f0/4
provider-dgavdadaev-sw-1(config-if)#sw mode trunk
provider-dgavdadaev-sw-1(config-if)#vlan 5
provider-dgavdadaev-sw-1(config-vlan)#name q42
provider-dgavdadaev-sw-1(config-vlan)#int vlan 5
provider-dgavdadaev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan5, changed state to up

provider-dgavdadaev-sw-1(config-if)#no sh
provider-dgavdadaev-sw-1(config-if)#vlan 6
provider-dgavdadaev-sw-1(config-vlan)#name sochi
provider-dgavdadaev-sw-1(config-vlan)#int vlan 6
provider-dgavdadaev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

provider-dgavdadaev-sw-1(config-if)#no sh
provider-dgavdadaev-sw-1(config-if)#exit
provider-dgavdadaev-sw-1(config)#exit
provider-dgavdadaev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

provider-dgavdadaev-sw-1#wr mem
Building configuration...
[OK]
```

Настройка подинтерфейсов на маршрутизаторе Донской

```
msk-donskaya-dgavdadaev-gw-1>en
Password:
msk-donskaya-dgavdadaev-gw-1#
msk-donskaya-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dgavdadaev-gw-1(config)#
msk-donskaya-dgavdadaev-gw-1(config)#int f0/1.5
msk-donskaya-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-donskaya-dgavdadaev-gw-1(config-subif)#enc dot1q 5
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip addr 10.128.255.1 255.255.255.252
msk-donskaya-dgavdadaev-gw-1(config-subif)#desc q42
msk-donskaya-dgavdadaev-gw-1(config-subif)#
msk-donskaya-dgavdadaev-gw-1(config-subif)#int f0/1.6
msk-donskaya-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.6, changed state to up

msk-donskaya-dgavdadaev-gw-1(config-subif)#enc dot1q 6
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip addr 10.128.255.5 255.255.255.252
msk-donskaya-dgavdadaev-gw-1(config-subif)#descr sochi
msk-donskaya-dgavdadaev-gw-1(config-subif)#exit
msk-donskaya-dgavdadaev-gw-1(config)#exit
msk-donskaya-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-dgavdadaev-gw-1#
```

Рис. 3: Настройка подинтерфейсов на маршрутизаторе Донской

Настройка маршрутизатора 42-го квартала

```
msk-q42-dgavdadaev-gw-1>en
Password:
msk-q42-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-dgavdadaev-gw-1(config)#
msk-q42-dgavdadaev-gw-1(config)#int f0/1
msk-q42-dgavdadaev-gw-1(config-if)#no sh

msk-q42-dgavdadaev-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-q42-dgavdadaev-gw-1(config-if)#int f0/1.5
msk-q42-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/1.5, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1.5, changed state to up

msk-q42-dgavdadaev-gw-1(config-subif)#enc dot1q 5
msk-q42-dgavdadaev-gw-1(config-subif)#ip addr 10.128.255.2 255.255.255.252
msk-q42-dgavdadaev-gw-1(config-subif)#descr donskaya
msk-q42-dgavdadaev-gw-1(config-subif)#exit
msk-q42-dgavdadaev-gw-1(config)#exit
msk-q42-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
msk-q42-dgavdadaev-gw-1#
```

Рис. 4: Настройка маршрутизатора сети 42-го квартала


```
sch-sochi-dgavdadaev-sw-1>en
Password:
sch-sochi-dgavdadaev-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-dgavdadaev-sw-1(config)#
sch-sochi-dgavdadaev-sw-1(config)#int f0/23
sch-sochi-dgavdadaev-sw-1(config-if)#sw mode trunk
sch-sochi-dgavdadaev-sw-1(config-if)#int f0/24
sch-sochi-dgavdadaev-sw-1(config-if)#sw mode trunk
sch-sochi-dgavdadaev-sw-1(config-if)#vlan 6
sch-sochi-dgavdadaev-sw-1(config-vlan)#name sochi
sch-sochi-dgavdadaev-sw-1(config-vlan)#int vlan 6
sch-sochi-dgavdadaev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan6, changed state to up

sch-sochi-dgavdadaev-sw-1(config-if)#no sh
sch-sochi-dgavdadaev-sw-1(config-if)#exit
sch-sochi-dgavdadaev-sw-1(config)#exit
sch-sochi-dgavdadaev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-dgavdadaev-sw-1#wr mem
Building configuration...
[OK]
sch-sochi-dgavdadaev-sw-1#
```

Рис. 5: Настройка коммутатора филиала в Сочи

Настройка маршрутизатора филиала в Сочи

```
sch-sochi-dgavdadaev-gw-1>en
Password:
sch-sochi-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-dgavdadaev-gw-1(config)#int f0/0
sch-sochi-dgavdadaev-gw-1(config-if)#no shut

sch-sochi-dgavdadaev-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

sch-sochi-dgavdadaev-gw-1(config-if)#int f0/0.6
sch-sochi-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.6, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.6, changed state to up

sch-sochi-dgavdadaev-gw-1(config-subif)#enc dot1q 6
sch-sochi-dgavdadaev-gw-1(config-subif)#ip addr 10.128.255.6 255.255.255.252
sch-sochi-dgavdadaev-gw-1(config-subif)#descr donskaya
sch-sochi-dgavdadaev-gw-1(config-subif)#exit
sch-sochi-dgavdadaev-gw-1(config)#exit
sch-sochi-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-dgavdadaev-gw-1#
```

Рис. 6: Настройка маршрутизатора филиала в Сочи

Настройка локальных сетей 42-го квартала

```
msk-q42-dgavdadaev-gw-1(config)#
msk-q42-dgavdadaev-gw-1(config)#int f0/0
msk-q42-dgavdadaev-gw-1(config-if)#no sh

msk-q42-dgavdadaev-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

msk-q42-dgavdadaev-gw-1(config-if)#int f0/0.201
msk-q42-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.201, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.201, changed state to up

msk-q42-dgavdadaev-gw-1(config-subif)#enc dot1q 201
msk-q42-dgavdadaev-gw-1(config-subif)#ip addr 10.129.0.1 255.255.255.0
msk-q42-dgavdadaev-gw-1(config-subif)#descr q42-main
msk-q42-dgavdadaev-gw-1(config-subif)#int f1/0
msk-q42-dgavdadaev-gw-1(config-if)#no sh

msk-q42-dgavdadaev-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

msk-q42-dgavdadaev-gw-1(config-if)#int f1/0.202
msk-q42-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet1/0.202, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0.202, changed state to up

msk-q42-dgavdadaev-gw-1(config-subif)#enc dot1q 202
msk-q42-dgavdadaev-gw-1(config-subif)#ip addr 10.129.1.1 255.255.255.0
msk-q42-dgavdadaev-gw-1(config-subif)#descr q42-management
msk-q42-dgavdadaev-gw-1(config-subif)#exit
msk-q42-dgavdadaev-gw-1(config)#exit
msk-q42-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```

Настройка коммутатора 42-го квартала

```
msk-q42-dgavdadaev-sw-1>en
Password:
msk-q42-dgavdadaev-sw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-q42-dgavdadaev-sw-1(config)#int f0/24
msk-q42-dgavdadaev-sw-1(config-if)#sw mode trunk

msk-q42-dgavdadaev-sw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

msk-q42-dgavdadaev-sw-1(config-if)#int f0/1
msk-q42-dgavdadaev-sw-1(config-if)#sw mode acc
msk-q42-dgavdadaev-sw-1(config-if)#sw acc vlan 201
% Access VLAN does not exist. Creating vlan 201
msk-q42-dgavdadaev-sw-1(config-if)#vlan 201
msk-q42-dgavdadaev-sw-1(config-vlan)#name q42-main
msk-q42-dgavdadaev-sw-1(config-vlan)#int vlan 201
msk-q42-dgavdadaev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan201, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan201, changed state to up

msk-q42-dgavdadaev-sw-1(config-if)#no sh
msk-q42-dgavdadaev-sw-1(config-if)#exit
msk-q42-dgavdadaev-sw-1(config)#exit
msk-q42-dgavdadaev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-dgavdadaev-sw-1#wr mem
Building configuration...
[OK]
msk-q42-dgavdadaev-sw-1#
```

Рис. 8: Настройка коммутатора основной сети 42-го квартала

Настройка сети общезития

```
msk-hostel-dgavdadaev-gw-1(config)#
msk-hostel-dgavdadaev-gw-1(config)#int g0/1
msk-hostel-dgavdadaev-gw-1(config-if)#sw trunk enc dot1q
msk-hostel-dgavdadaev-gw-1(config-if)#sw mode trunk

msk-hostel-dgavdadaev-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

msk-hostel-dgavdadaev-gw-1(config-if)#int f0/1
msk-hostel-dgavdadaev-gw-1(config-if)#sw trunk enc dot1q
msk-hostel-dgavdadaev-gw-1(config-if)#sw mode trunk

msk-hostel-dgavdadaev-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

msk-hostel-dgavdadaev-gw-1(config-if)#vlan 202
msk-hostel-dgavdadaev-gw-1(config-vlan)#name q42-management
msk-hostel-dgavdadaev-gw-1(config-vlan)#int vlan 202
msk-hostel-dgavdadaev-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan202, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan202, changed state to up

msk-hostel-dgavdadaev-gw-1(config-if)#no sh
msk-hostel-dgavdadaev-gw-1(config-if)#ip addr 10.129.1.2 255.255.255.0
msk-hostel-dgavdadaev-gw-1(config-if)#
msk-hostel-dgavdadaev-gw-1(config-if)#vlan 301
msk-hostel-dgavdadaev-gw-1(config-vlan)#name hostel-main
msk-hostel-dgavdadaev-gw-1(config-vlan)#int vlan 301
msk-hostel-dgavdadaev-gw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-dgavdadaev-gw-1(config-if)#no sh
msk-hostel-dgavdadaev-gw-1(config-if)#ip addr 10.129.128.1 255.255.255.0
msk-hostel-dgavdadaev-gw-1(config-if)#exit
msk-hostel-dgavdadaev-gw-1(config)#exit
```

Настройка коммутатора сети общежития

```
msk-hostel-dgavdadaev-sw-1>en
Password:
msk-hostel-dgavdadaev-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-hostel-dgavdadaev-sw-1(config)#
msk-hostel-dgavdadaev-sw-1(config)#int g0/1
msk-hostel-dgavdadaev-sw-1(config-if)#sw mode trunk
msk-hostel-dgavdadaev-sw-1(config-if)#int f0/1
msk-hostel-dgavdadaev-sw-1(config-if)#sw mode acc
msk-hostel-dgavdadaev-sw-1(config-if)#sw acc vlan 301
% Access VLAN does not exist. Creating vlan 301
msk-hostel-dgavdadaev-sw-1(config-if)#vlan 301
msk-hostel-dgavdadaev-sw-1(config-vlan)#name hostel-main
msk-hostel-dgavdadaev-sw-1(config-vlan)#int vlan 301
msk-hostel-dgavdadaev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan301, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan301, changed state to up

msk-hostel-dgavdadaev-sw-1(config-if)#no sh
msk-hostel-dgavdadaev-sw-1(config-if)#exit
msk-hostel-dgavdadaev-sw-1(config)#exit
msk-hostel-dgavdadaev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-hostel-dgavdadaev-sw-1#wr mem
Building configuration...
[OK]
msk-hostel-dgavdadaev-sw-1#
```

Рис. 10: Настройка коммутатора сети общежития

Настройка локальных сетей филиала в Сочи

```
sch-sochi-dgavdadaev-gw-1(config)#
sch-sochi-dgavdadaev-gw-1(config)#int f0/0.401
sch-sochi-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.401, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

sch-sochi-dgavdadaev-gw-1(config-subif)#enc dot1q 401
sch-sochi-dgavdadaev-gw-1(config-subif)#ip addr 10.130.0.1 255.255.255.0
sch-sochi-dgavdadaev-gw-1(config-subif)#desc sochi-management
sch-sochi-dgavdadaev-gw-1(config-subif)#
sch-sochi-dgavdadaev-gw-1(config-subif)#int f0/0.402
sch-sochi-dgavdadaev-gw-1(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.402, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.402, changed state to up

sch-sochi-dgavdadaev-gw-1(config-subif)#enc dot1q 402
sch-sochi-dgavdadaev-gw-1(config-subif)#ip addr 10.130.1.1 255.255.255.0
sch-sochi-dgavdadaev-gw-1(config-subif)#descr sochi-management
sch-sochi-dgavdadaev-gw-1(config-subif)#
sch-sochi-dgavdadaev-gw-1(config-subif)#int f0/0.401
sch-sochi-dgavdadaev-gw-1(config-subif)#no desc sochi-management
sch-sochi-dgavdadaev-gw-1(config-subif)#descr sochi-main
sch-sochi-dgavdadaev-gw-1(config-subif)#exit
sch-sochi-dgavdadaev-gw-1(config)#exit
sch-sochi-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
sch-sochi-dgavdadaev-gw-1#
```

Рис. 11: Настройка локальных VLAN на маршрутизаторе филиала в Сочи

Настройка коммутатора филиала в Сочи

```
sch-sochi-dgavdadaev-sw-1>en
Password:
sch-sochi-dgavdadaev-sw-1#
sch-sochi-dgavdadaev-sw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
sch-sochi-dgavdadaev-sw-1(config)#
sch-sochi-dgavdadaev-sw-1(config)#int f0/1
sch-sochi-dgavdadaev-sw-1(config-if)#sw mode acc
sch-sochi-dgavdadaev-sw-1(config-if)#sw acc vlan 401
% Access VLAN does not exist. Creating vlan 401
sch-sochi-dgavdadaev-sw-1(config-if)#vlan 401
sch-sochi-dgavdadaev-sw-1(config-vlan)#name sochi-main
sch-sochi-dgavdadaev-sw-1(config-vlan)#int vlan 401
sch-sochi-dgavdadaev-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

sch-sochi-dgavdadaev-sw-1(config-if)#no sh
sch-sochi-dgavdadaev-sw-1(config-if)#exit
sch-sochi-dgavdadaev-sw-1(config)#exit
sch-sochi-dgavdadaev-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

sch-sochi-dgavdadaev-sw-1#wr mem'
      ^
% Invalid input detected at '^' marker.

sch-sochi-dgavdadaev-sw-1#wr mem
Building configuration...
[OK]
```


Настройка статических маршрутов на Донской

```
msk-donskaya-dgavdadaev-gw-1>en
Password:
msk-donskaya-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-dgavdadaev-gw-1(config)#
msk-donskaya-dgavdadaev-gw-1(config)#ip route 10.129.0.0 255.255.0.0 10.128.255.2
msk-donskaya-dgavdadaev-gw-1(config)#ip route 10.130.0.0 255.255.0.0 10.128.255.6
msk-donskaya-dgavdadaev-gw-1(config)#int f0/1.5
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip nat inside
msk-donskaya-dgavdadaev-gw-1(config-subif)#int f0/1.6
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip nat inside
msk-donskaya-dgavdadaev-gw-1(config-subif)#
msk-donskaya-dgavdadaev-gw-1(config-subif)#ip access-list ext nat-inet
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark q42
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit ip host 10.129.0.200 any
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit ip host 10.129.128.200 any
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#remark sochi
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#permit ip host 10.130.0.200 any
msk-donskaya-dgavdadaev-gw-1(config-ext-nacl)#exit
msk-donskaya-dgavdadaev-gw-1(config)#exit
msk-donskaya-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-donskaya-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
msk-donskaya-dgavdadaev-gw-1#
```

Рис. 13: Настройка статических маршрутов и NAT на маршрутизаторе Донской

Настройка маршрутизации на 42-м квартале

```
msk-q42-dgavdadaev-gw-1>en
Password:
msk-q42-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
msk-q42-dgavdadaev-gw-1(config)#
msk-q42-dgavdadaev-gw-1(config)#
msk-q42-dgavdadaev-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.1
msk-q42-dgavdadaev-gw-1(config)#ip route 10.129.128.0 255.255.128.0 10.129.1.2
msk-q42-dgavdadaev-gw-1(config)#exit
msk-q42-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

msk-q42-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
msk-q42-dgavdadaev-gw-1#
```

Рис. 14: Настройка статической маршрутизации на маршрутизаторе 42-го квартала

Настройка маршрутизации в сети общежития

```
mik-hostel-dgavdadaev-gw-1#
mik-hostel-dgavdadaev-gw-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
mik-hostel-dgavdadaev-gw-1(config)#
mik-hostel-dgavdadaev-gw-1(config)#ip routing
mik-hostel-dgavdadaev-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.129.1.1
mik-hostel-dgavdadaev-gw-1(config)#exit
mik-hostel-dgavdadaev-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

mik-hostel-dgavdadaev-gw-1#wr mem
Building configuration...
[OK]
mik-hostel-dgavdadaev-gw-1#
```

Рис. 15: Настройка маршрутизации на устройстве сети общежития

```
sch-sochi-dgavdadaev-gw-1#  
sch-sochi-dgavdadaev-gw-1#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
sch-sochi-dgavdadaev-gw-1(config)#ip route 0.0.0.0 0.0.0.0 10.128.255.5  
sch-sochi-dgavdadaev-gw-1(config)#exit  
sch-sochi-dgavdadaev-gw-1#  
%SYS-5-CONFIG_I: Configured from console by console  
  
sch-sochi-dgavdadaev-gw-1#wr mem  
Building configuration...  
[OK]  
sch-sochi-dgavdadaev-gw-1#
```

Рис. 16: Настройка маршрута по умолчанию на маршрутизаторе филиала в Сочи

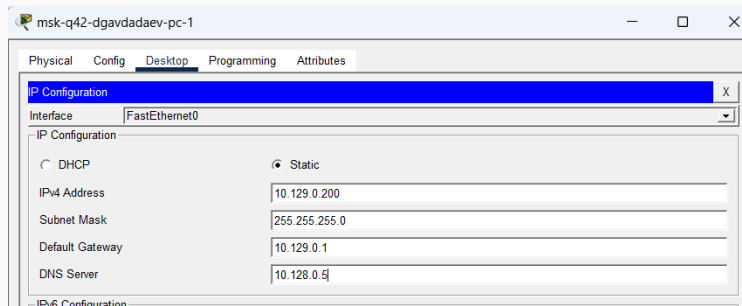


Рис. 17: Настройка IP-адресации рабочей станции 42-го квартала

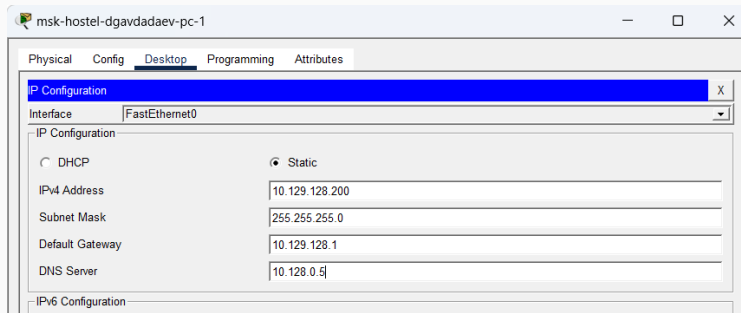


Рис. 18: Настройка IP-адресации рабочей станции сети общезнания

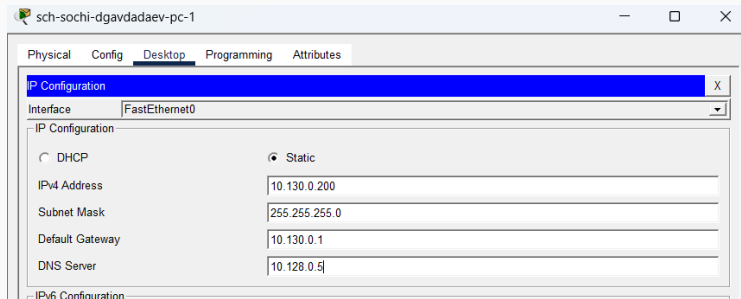


Рис. 19: Настройка IP-адресации рабочей станции филиала в Сочи

Проверка работоспособности

admin-donskaya-dgavdadaev-1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.129.128.200

Pinging 10.129.128.200 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.129.128.200: bytes=32 time<1ms TTL=125
Reply from 10.129.128.200: bytes=32 time<1ms TTL=125

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.130.0.200

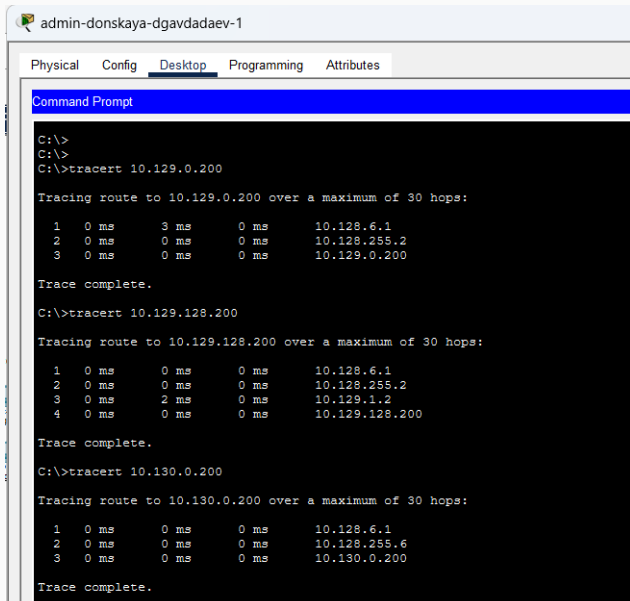
Pinging 10.130.0.200 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.130.0.200: bytes=32 time=8ms TTL=126
Reply from 10.130.0.200: bytes=32 time=4ms TTL=126

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 8ms, Average = 6ms

C:\>
```

Трассировка маршрутов с Донской



The screenshot shows a Windows desktop with a taskbar at the top containing icons for a folder, a web browser, and a file explorer. The active window is titled "admin-donskaya-dgavdadaev-1" and has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is selected, and a "Command Prompt" window is open. The Command Prompt displays the results of three traceroute commands executed from the C:\> directory.

```
C:\>
C:\>
C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  1  0 ms    3 ms    0 ms    10.128.6.1
  2  0 ms    0 ms    0 ms    10.128.255.2
  3  0 ms    0 ms    0 ms    10.129.0.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.128.6.1
  2  0 ms    0 ms    0 ms    10.128.255.2
  3  0 ms    2 ms    0 ms    10.129.1.2
  4  0 ms    0 ms    0 ms    10.129.128.200

Trace complete.

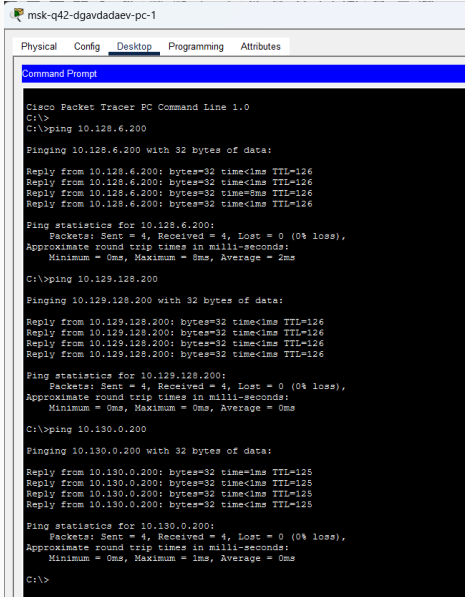
C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.128.6.1
  2  0 ms    0 ms    0 ms    10.128.255.6
  3  0 ms    0 ms    0 ms    10.130.0.200

Trace complete.
```

Проверка связи с компьютера 42-го квартала



```
msk-q42-dgavdadaev-pc-1
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.128.6.200

Pinging 10.128.6.200 with 32 bytes of data:

Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time=8ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.128.6.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 10.129.128.200

Pinging 10.129.128.200 with 32 bytes of data:

Reply from 10.129.128.200: bytes=32 time<1ms TTL=126
Reply from 10.129.128.200: bytes=32 time<1ms TTL=126
Reply from 10.129.128.200: bytes=32 time<1ms TTL=126
Reply from 10.129.128.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.130.0.200

Pinging 10.130.0.200 with 32 bytes of data:

Reply from 10.130.0.200: bytes=32 time=1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125
Reply from 10.130.0.200: bytes=32 time<1ms TTL=125

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Трассировка с компьютера 42-го квартала

msk-q42-dgavdadaev-pc-1

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>tracert 10.128.6.200

Tracing route to 10.128.6.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.0.1
  2  0 ms    0 ms    0 ms    10.128.255.1
  3  0 ms    1 ms    0 ms    10.128.6.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.0.1
  2  0 ms    0 ms    0 ms    10.129.1.2
  3  0 ms    0 ms    0 ms    10.129.128.200

Trace complete.

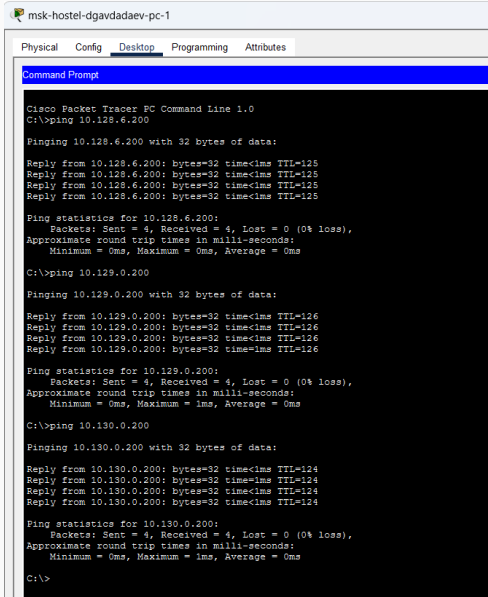
C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.0.1
  2  0 ms    0 ms    0 ms    10.128.255.1
  3  1 ms    4 ms    0 ms    10.128.255.6
  4  0 ms    0 ms    0 ms    10.130.0.200

Trace complete.
```

Проверка связи с компьютера общежития



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'msk-hostel-dgavdadaev-pc-1'. The window has tabs for Physical, Config, Desktop (selected), Programming, and Attributes. The Command Prompt shows the following commands and output:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.6.200

Pinging 10.128.6.200 with 32 bytes of data:

Reply from 10.128.6.200: bytes=32 time<1ms TTL=125
Reply from 10.128.6.200: bytes=32 time<1ms TTL=125
Reply from 10.128.6.200: bytes=32 time<1ms TTL=125
Reply from 10.128.6.200: bytes=32 time<1ms TTL=125

Ping statistics for 10.128.6.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126
Reply from 10.129.0.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.130.0.200

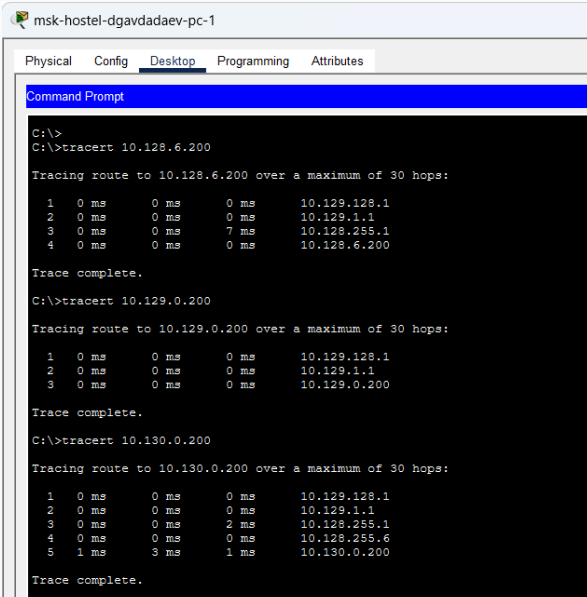
Pinging 10.130.0.200 with 32 bytes of data:

Reply from 10.130.0.200: bytes=32 time<1ms TTL=124
Reply from 10.130.0.200: bytes=32 time<1ms TTL=124
Reply from 10.130.0.200: bytes=32 time<1ms TTL=124
Reply from 10.130.0.200: bytes=32 time<1ms TTL=124

Ping statistics for 10.130.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Трассировка с компьютера общежития



msk-hostel-dgavdadaev-pc-1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>
C:\>tracert 10.128.6.200

Tracing route to 10.128.6.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.128.1
  2  0 ms    0 ms    0 ms    10.129.1.1
  3  0 ms    0 ms    7 ms    10.128.255.1
  4  0 ms    0 ms    0 ms    10.128.6.200

Trace complete.

C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.128.1
  2  0 ms    0 ms    0 ms    10.129.1.1
  3  0 ms    0 ms    0 ms    10.129.0.200

Trace complete.

C:\>tracert 10.130.0.200

Tracing route to 10.130.0.200 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.129.128.1
  2  0 ms    0 ms    0 ms    10.129.1.1
  3  0 ms    0 ms    2 ms    10.128.255.1
  4  0 ms    0 ms    0 ms    10.128.255.6
  5  1 ms    3 ms    1 ms    10.130.0.200

Trace complete.
```

Проверка связи с компьютера филиала в Сочи

```
sch-sochi-dgavdadaev-pc-1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.129.0.200

Pinging 10.129.0.200 with 32 bytes of data:

Reply from 10.129.0.200: bytes=32 time<1ms TTL=125
Reply from 10.129.0.200: bytes=32 time<1ms TTL=125
Reply from 10.129.0.200: bytes=32 time<1ms TTL=125
Reply from 10.129.0.200: bytes=32 time=8ms TTL=125

Ping statistics for 10.129.0.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping 10.129.128.200

Pinging 10.129.128.200 with 32 bytes of data:

Reply from 10.129.128.200: bytes=32 time<1ms TTL=124
Reply from 10.129.128.200: bytes=32 time<1ms TTL=124
Reply from 10.129.128.200: bytes=32 time<1ms TTL=124
Reply from 10.129.128.200: bytes=32 time<1ms TTL=124

Ping statistics for 10.129.128.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.128.6.200

Pinging 10.128.6.200 with 32 bytes of data:

Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time=7ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126
Reply from 10.128.6.200: bytes=32 time<1ms TTL=126

Ping statistics for 10.128.6.200:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>
```


Трассировка с компьютера филиала в Сочи

sch-sochi-dgavdadaev-pc-1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>tracert 10.129.0.200

Tracing route to 10.129.0.200 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.130.0.1
  1  0 ms    2 ms    0 ms    10.128.255.5
  2  1 ms    0 ms    0 ms    10.128.255.2
  3  0 ms    0 ms    0 ms    10.129.0.200

Trace complete.

C:\>tracert 10.129.128.200

Tracing route to 10.129.128.200 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.130.0.1
  1  0 ms    0 ms    0 ms    10.128.255.5
  2  0 ms    0 ms    0 ms    10.128.255.2
  3  0 ms    4 ms    0 ms    10.129.1.2
  4  0 ms    0 ms    0 ms    10.129.128.200

Trace complete.

C:\>tracert 10.128.6.200

Tracing route to 10.128.6.200 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    10.130.0.1
  1  0 ms    0 ms    0 ms    10.128.255.5
  2  0 ms    0 ms    0 ms    10.128.6.200

Trace complete.
```

Проверка таблиц маршрутизации

Таблица маршрутизации Донской

```
msh-donskaya-dgavdadaev-gw-1#
msh-donskaya-dgavdadaev-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 18 subnets, 4 masks
C    10.128.0.0/24 is directly connected, FastEthernet0/0.3
L    10.128.0.1/32 is directly connected, FastEthernet0/0.3
C    10.128.1.0/24 is directly connected, FastEthernet0/0.2
L    10.128.1.1/32 is directly connected, FastEthernet0/0.2
C    10.128.3.0/24 is directly connected, FastEthernet0/0.101
L    10.128.3.1/32 is directly connected, FastEthernet0/0.101
C    10.128.4.0/24 is directly connected, FastEthernet0/0.102
L    10.128.4.1/32 is directly connected, FastEthernet0/0.102
C    10.128.5.0/24 is directly connected, FastEthernet0/0.103
L    10.128.5.1/32 is directly connected, FastEthernet0/0.103
C    10.128.6.0/24 is directly connected, FastEthernet0/0.104
L    10.128.6.1/32 is directly connected, FastEthernet0/0.104
C    10.128.255.0/30 is directly connected, FastEthernet0/1.5
L    10.128.255.1/32 is directly connected, FastEthernet0/1.5
C    10.128.255.4/30 is directly connected, FastEthernet0/1.6
L    10.128.255.5/32 is directly connected, FastEthernet0/1.6
S    10.129.0.0/16 [1/0] via 10.128.255.2
S    10.130.0.0/16 [1/0] via 10.128.255.6
198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    198.51.100.0/28 is directly connected, FastEthernet0/1.4
L    198.51.100.2/32 is directly connected, FastEthernet0/1.4
S*   0.0.0.0/0 [1/0] via 198.51.100.1
```

Таблица маршрутизации 42-го квартала

```
msk-q42-dgavdadaev-gw-1#
msk-q42-dgavdadaev-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.1 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 7 subnets, 4 masks
C       10.128.255.0/30 is directly connected, FastEthernet0/1.5
L       10.128.255.2/32 is directly connected, FastEthernet0/1.5
C       10.129.0.0/24 is directly connected, FastEthernet0/0.201
L       10.129.0.1/32 is directly connected, FastEthernet0/0.201
C       10.129.1.0/24 is directly connected, FastEthernet1/0.202
L       10.129.1.1/32 is directly connected, FastEthernet1/0.202
S       10.129.128.0/17 [1/0] via 10.129.1.2
S*    0.0.0.0/0 [1/0] via 10.128.255.1
```

Рис. 29: Проверка таблицы маршрутизации на маршрутизаторе 42-го квартала

```
msk-hostel-dgavdadaev-gw-1#
msk-hostel-dgavdadaev-gw-1#sh ip rou
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.129.1.1 to network 0.0.0.0

    10.0.0.0/24 is subnetted, 2 subnets
C       10.129.1.0 is directly connected, Vlan202
C       10.129.128.0 is directly connected, Vlan301
S*    0.0.0.0/0 [1/0] via 10.129.1.1
```

Рис. 30: Проверка таблицы маршрутизации на устройстве сети общежития

Таблица маршрутизации филиала в Сочи

```
sch-sochi-dgavdadaev-gw-1#
sch-sochi-dgavdadaev-gw-1#sh ip rou
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.128.255.5 to network 0.0.0.0

    10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
C       10.128.255.4/30 is directly connected, FastEthernet0/0.6
L       10.128.255.6/32 is directly connected, FastEthernet0/0.6
C       10.130.0.0/24 is directly connected, FastEthernet0/0.401
L       10.130.0.1/32 is directly connected, FastEthernet0/0.401
C       10.130.1.0/24 is directly connected, FastEthernet0/0.402
L       10.130.1.1/32 is directly connected, FastEthernet0/0.402
S*    0.0.0.0/0 [1/0] via 10.128.255.5
```

Рис. 31: Проверка таблицы маршрутизации на маршрутизаторе филиала в Сочи

Выводы

В ходе выполнения лабораторной работы:

- построена топология сети в Cisco Packet Tracer
- настроены VLAN и trunk-соединения
- созданы подинтерфейсы с инкапсуляцией **dot1q**
- назначены IP-адреса маршрутизаторам и рабочим станциям
- настроена статическая маршрутизация между площадками
- настроены маршруты по умолчанию
- проверена доступность удалённых сетей
- просмотрены таблицы маршрутизации на основных устройствах